

## SAFETY DATA SHEET

Creation Date 26-October-2009

Revision Date 14-January-2025

Revision Number 3

### 1. Identification

**Product Name** Guanidine thiocyanate

**Cat No. :** S60106

**CAS-No** 593-84-0

**Synonyms** Guanidinium isothiocyanate; Thiocyanic acid, compound with Guanidine (1:1)

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                |              |
|------------------------------------------------|--------------|
| <b>Acute oral toxicity</b>                     | Category 4   |
| <b>Acute dermal toxicity</b>                   | Category 4   |
| <b>Acute Inhalation Toxicity</b>               | Category 4   |
| <b>Skin Corrosion/Irritation</b>               | Category 1 C |
| <b>Serious Eye Damage/Eye Irritation</b>       | Category 1   |
| <b>Health Hazards Not Otherwise Classified</b> | Category 1   |
| Contact with acids liberates very toxic gas    |              |
| Corrosive to the respiratory tract             |              |

#### Label Elements

##### **Signal Word**

Danger

**Hazard Statements**

Harmful if swallowed, in contact with skin or if inhaled  
Causes severe skin burns and eye damage  
Contact with acids liberates very toxic gas  
Corrosive to the respiratory tract

**Precautionary Statements****Prevention**

Take any precaution to avoid mixing with acids  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wear respiratory protection  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Immediately call a POISON CENTER/doctor  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Rinse mouth  
Do NOT induce vomiting  
Wash contaminated clothing before reuse

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Harmful to aquatic life with long lasting effects  
Light sensitive

### 3. Composition/Information on Ingredients

| Component                                      | CAS-No   | Weight % |
|------------------------------------------------|----------|----------|
| Thiocyanic acid, compound with guanidine (1:1) | 593-84-0 | >95      |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required. Keep eye wide open while rinsing.

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Call a physician immediately.

|                                        |                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Call a physician or poison control center immediately. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. |
| <b>Ingestion</b>                       | Immediate medical attention is required. Do NOT induce vomiting. Drink plenty of water. Never give anything by mouth to an unconscious person.                                                                                                                                                                                               |
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation                                              |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                        |

## 5. Fire-fighting measures

|                                         |                                                                   |
|-----------------------------------------|-------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                          |
| <b>Flash Point</b>                      | No information available                                          |
| <b>Method -</b>                         | No information available                                          |
| <b>Autoignition Temperature</b>         | Not applicable                                                    |
| <b>Explosion Limits</b>                 |                                                                   |
| <b>Upper</b>                            | No data available                                                 |
| <b>Lower</b>                            | No data available                                                 |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                          |
| <b>Sensitivity to Static Discharge</b>  | No information available                                          |

### Specific Hazards Arising from the Chemical

The product causes burns of eyes, skin and mucous membranes.

### Hazardous Combustion Products

Hydrogen cyanide (hydrocyanic acid). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Sulfur oxides.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

**Health**  
4

**Flammability**  
1

**Instability**  
1

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                  |                                                                                                                                                                  |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.                                      |
| <b>Environmental Precautions</b> | Should not be released into the environment. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system. |

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from light.                                                                                                                                   |

Corrosives area. Keep under nitrogen. Incompatible Materials. Acids. Strong oxidizing agents.

## 8. Exposure controls / personal protection

### Exposure Guidelines

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.  
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### **Eye Protection** **Hand Protection**

Goggles  
Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

Prevent product from entering drains.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                            |                                   |
|----------------------------|-----------------------------------|
| <b>Physical State</b>      | Powder Solid                      |
| <b>Appearance</b>          | Off-white                         |
| <b>Odor</b>                | Odorless                          |
| <b>Odor Threshold</b>      | No information available          |
| <b>pH</b>                  | approx 4.8 - 6.0 20% aq. solution |
| <b>Melting Point/Range</b> | 118 - 122 °C / 244.4 - 251.6 °F   |
| <b>Boiling Point/Range</b> | No information available          |
| <b>Flash Point</b>         | No information available          |
| <b>Evaporation Rate</b>    | Not applicable                    |

|                                        |                          |
|----------------------------------------|--------------------------|
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | negligible               |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | No information available |
| Solubility                             | Soluble in water         |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | Not applicable           |
| Decomposition Temperature              | No information available |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | C2 H6 N4 S               |
| Molecular Weight                       | 118.16                   |

## 10. Stability and reactivity

|                                  |                                                                                                                                                 |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                                                                             |
| Stability                        | Light sensitive.                                                                                                                                |
| Conditions to Avoid              | Avoid dust formation. Incompatible products. Exposure to light. Excess heat.                                                                    |
| Incompatible Materials           | Acids, Strong oxidizing agents                                                                                                                  |
| Hazardous Decomposition Products | Hydrogen cyanide (hydrocyanic acid), Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Nitrogen oxides (NO <sub>x</sub> ), Sulfur oxides |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                                        |
| Hazardous Reactions              | None under normal processing.                                                                                                                   |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

| Component                                      | LD50 Oral                          | LD50 Dermal | LC50 Inhalation                                                                                   |
|------------------------------------------------|------------------------------------|-------------|---------------------------------------------------------------------------------------------------|
| Thiocyanic acid, compound with guanidine (1:1) | LD50 = 593 mg/kg (rat)<br>OECD 401 | Not listed  | LC50 combined: 5.319 mg/L<br>LC50 Males: 7.655 mg/L<br>LC50 Females: 3.181 mg/L<br>(Rat) OECD 403 |

**Toxicologically Synergistic Products** No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Causes burns by all exposure routes                                                      |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                                      | CAS-No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------------------------------------|----------|------------|------------|------------|------------|------------|
| Thiocyanic acid, compound with guanidine (1:1) | 593-84-0 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

|                                                   |                                                                                                                                                                                                                                                            |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Developmental Effects</b>                      | No information available.                                                                                                                                                                                                                                  |
| <b>Teratogenicity</b>                             | No information available.                                                                                                                                                                                                                                  |
| <b>STOT - single exposure</b>                     | None known                                                                                                                                                                                                                                                 |
| <b>STOT - repeated exposure</b>                   | None known                                                                                                                                                                                                                                                 |
| <b>Aspiration hazard</b>                          | No information available                                                                                                                                                                                                                                   |
| <b>Symptoms / effects, both acute and delayed</b> | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                                                                                                                                                                   |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                                                                                                                                             |

## 12. Ecological information

### Ecotoxicity

Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component                                      | Freshwater Algae | Freshwater Fish                            | Microtox   | Water Flea         |
|------------------------------------------------|------------------|--------------------------------------------|------------|--------------------|
| Thiocyanic acid, compound with guanidine (1:1) | Not listed       | Poecilia reticulata:<br>LC50=89.1 mg/L 96h | Not listed | EC50=42.4 mg/L 48h |

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

| Component                                      | log Pow |
|------------------------------------------------|---------|
| Thiocyanic acid, compound with guanidine (1:1) | -1.38   |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

|                             |                                                |
|-----------------------------|------------------------------------------------|
| <b>UN-No</b>                | UN3261                                         |
| <b>Proper Shipping Name</b> | Corrosive solid, acidic, organic, n.o.s.       |
| <b>Technical Name</b>       | Thiocyanic acid, compound with guanidine (1:1) |
| <b>Hazard Class</b>         | 8                                              |
| <b>Packing Group</b>        | III                                            |

### TDG

|                             |                                          |
|-----------------------------|------------------------------------------|
| <b>UN-No</b>                | UN3261                                   |
| <b>Proper Shipping Name</b> | Corrosive solid, acidic, organic, n.o.s. |
| <b>Hazard Class</b>         | 8                                        |
| <b>Packing Group</b>        | III                                      |

### IATA

|                             |                                          |
|-----------------------------|------------------------------------------|
| <b>UN-No</b>                | UN3261                                   |
| <b>Proper Shipping Name</b> | Corrosive solid, acidic, organic, n.o.s. |
| <b>Hazard Class</b>         | 8                                        |
| <b>Packing Group</b>        | III                                      |

### IMDG/IMO

|                             |                                          |
|-----------------------------|------------------------------------------|
| <b>UN-No</b>                | UN3261                                   |
| <b>Proper Shipping Name</b> | Corrosive solid, acidic, organic, n.o.s. |

Hazard Class 8  
Packing Group III

## 15. Regulatory information

### International Inventories

| Component                                      | CAS-No   | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|------------------------------------------------|----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Thiocyanic acid, compound with guanidine (1:1) | 593-84-0 | X   | -    | X    | ACTIVE                                        | 209-812-1 | -      | -   |

| Component                                      | CAS-No   | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|------------------------------------------------|----------|-------|------|------|------|------|------|-------|-------|
| Thiocyanic acid, compound with guanidine (1:1) | 593-84-0 | X     | -    | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

### Other International Regulations

Authorisation/Restrictions according to EU REACH Not applicable

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                                      | CAS-No   | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------------------------------------|----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Thiocyanic acid, compound with guanidine (1:1) | 593-84-0 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                      | CAS-No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------------------------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Thiocyanic acid, compound with guanidine (1:1) | 593-84-0 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

#### Prepared By

Product Safety Department  
Email: [chem.techinfo@thermofisher.com](mailto:chem.techinfo@thermofisher.com)  
[www.thermofisher.com](http://www.thermofisher.com)

|                         |                                                                                                                                                                                    |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Creation Date</b>    | 26-October-2009                                                                                                                                                                    |
| <b>Revision Date</b>    | 14-January-2025                                                                                                                                                                    |
| <b>Print Date</b>       | 14-January-2025                                                                                                                                                                    |
| <b>Revision Summary</b> | This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals. |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**